

Data_Analysis_LE

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We decided that we wanted to model the probability of receiving professional sanctions based on the other variables available in the dataset.

```
library(readr)
library(ggplot2)
hallucination_cases <- read_csv("hallucination_cases.csv")

## Rows: 2078 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr  (17): Case Name, Court, State(s), Party(ies), AI Tool, Hallucination It...
## date  (1): Date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
#checking types of parties
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
hallucination_cases %>%
  count(`Party(ies)`)
```

```
## # A tibble: 17 x 2
##   'Party(ies)'      n
##   <chr>           <int>
## 1 Arbitrator         1
## 2 Expert            15
## 3 Federal Defender   1
```

```
## 4 Gouvernement Lawyer      3
## 5 Judge                    26
## 6 Judge; Lawyer            2
## 7 Judge; Prosecutor         1
## 8 Lawyer                   819
## 9 Lawyer; Judge            2
## 10 Lawyer; Pro Se Litigant  1
## 11 Paralegal               1
## 12 Paralegal; Lawyer        1
## 13 Pro Se Litigant         1194
## 14 Pro Se Litigant; Lawyer  1
## 15 Prosecutor               3
## 16 Prosecutor; Judge        1
## 17 <NA>                     6
```

```
#checking legal fields
hallucination_cases %>%
  count(`Legal Field Primary`)
```

```
## # A tibble: 15 x 2
##   `Legal Field Primary`     n
##   <chr>                  <int>
## 1 IP                      23
## 2 administrative         258
## 3 bankruptcy             54
## 4 civil rights           203
## 5 contract               450
## 6 criminal                64
## 7 employment             189
## 8 family                  136
## 9 habeas                  10
## 10 immigration            36
## 11 landlord-tenant        16
## 12 other                  173
## 13 tax                    24
## 14 tort                   181
## 15 <NA>                   261
```

```
#checking what the distribution of professional sanction vs no professional sanction
hallucination_cases %>%
  count(`Professional Sanction`)
```

```
## # A tibble: 2 x 2
##   `Professional Sanction`     n
##   <chr>                  <int>
## 1 No                      1915
## 2 Yes                     163
```

```
#checking what the distribution of professional sanction vs no professional sanction
hallucination_cases %>%
  count(`State(s)`)
```

```
## # A tibble: 45 x 2
```

```
##   'State(s)'          n
##   <chr>              <int>
## 1 Argentina          8
## 2 Australia          112
## 3 Austria            2
## 4 Belgium            8
## 5 Brazil             41
## 6 Canada             217
## 7 Canada; Domestic Arbitration 1
## 8 Chile              3
## 9 China              1
## 10 Colombia          3
## # i 35 more rows
```

Because we can see that there a lot of countries with only one case, it doesn't make sense to make every country a variable. That would add 45 predictors to the model. We will thus make a variable for all states with 50 or more cases and other: so Australia, Canada, Israel, UK, the US and all other countries.

```
hallucination_cases <- hallucination_cases |>
  filter(!is.na(`State(s)`) |>
    mutate(
      country_group = if_else(
        `State(s)` %in% c("Australia", "Canada", "Israel", "UK", "United States"),
        `State(s)`,
        "Other"
      )
    )
  )
```

Looking at the where we get the data from, we can see that the variable AI tool contains “implied” which is when AI usage was evident but not admitted.

```
#checking what the distribution of professional sanction vs no professional sanction
hallucination_cases %>%
  count(`AI Tool`)
```

```
## # A tibble: 103 x 2
##   'AI Tool'          n
##   <chr>              <int>
## 1 "\"Internet searches\"" 1
## 2 "Amicus (Casemine)"    1
## 3 "Archie (Smokeball)"  1
## 4 "Athena AI"           1
## 5 "Bard"                 1
## 6 "Bloomberg Law"       1
## 7 "CHATGPT 3"           1
## 8 "Centient AI"         2
## 9 "Cetient"             1
## 10 "ChatGPT"            113
## # i 93 more rows
```

Similarly to country, but more extreme, we have 104 options for AI tool, which is too many variables. We can thus separate this into three categories: Implied, Unidentified, and Specific AI Tool.

```

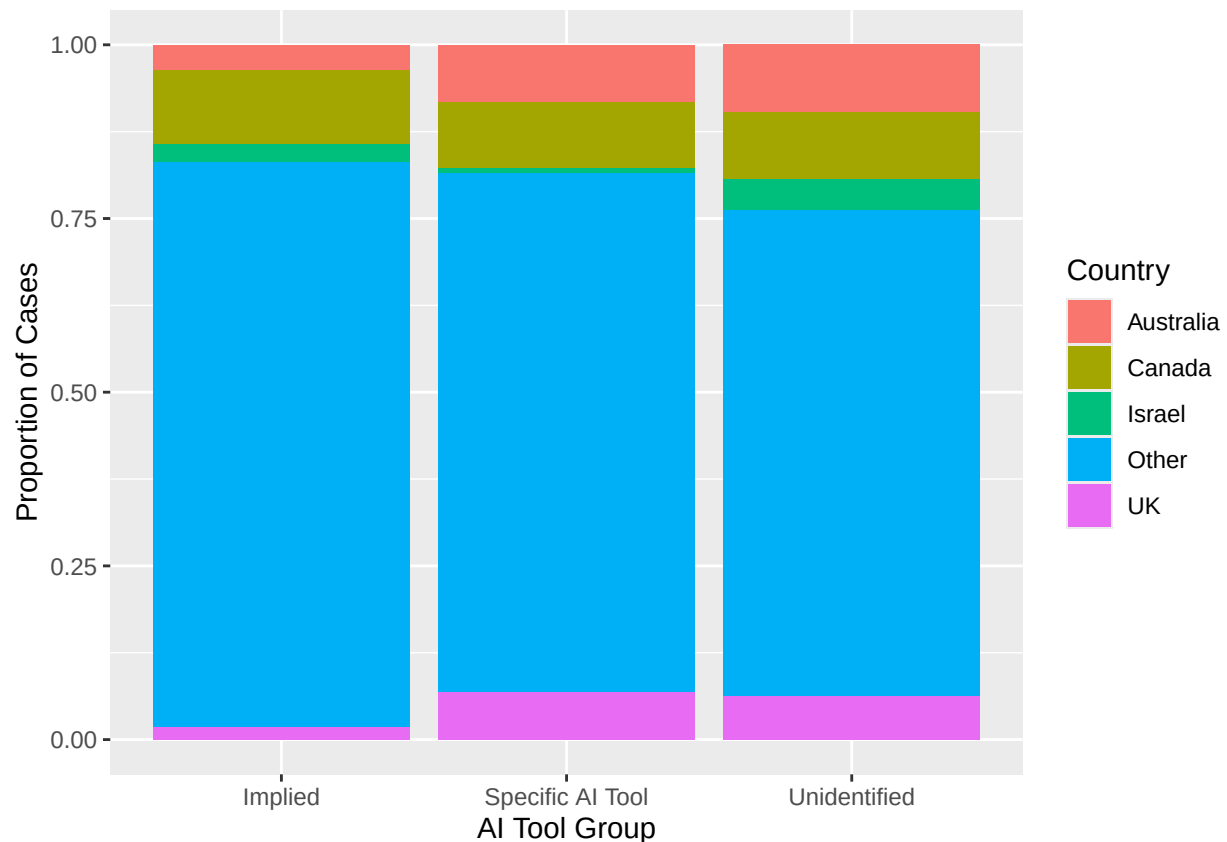
hallucination_cases <- hallucination_cases |>
  filter(!is.na(`AI Tool`)) %>%
  mutate(
    ai_tool_group = case_when(
      `AI Tool` == "Unidentified" ~ "Unidentified",
      `AI Tool` == "Implied" ~ "Implied",
      `AI Tool` == "implied" ~ "Implied",
      TRUE ~ "Specific AI Tool"
    )
  )

```

```

hallucination_cases |>
  ggplot(aes(x=ai_tool_group, fill = country_group)) +
  geom_bar(position = "fill") +
  labs(x = "AI Tool Group",
       y = "Proportion of Cases",
       fill = "Country")

```



```

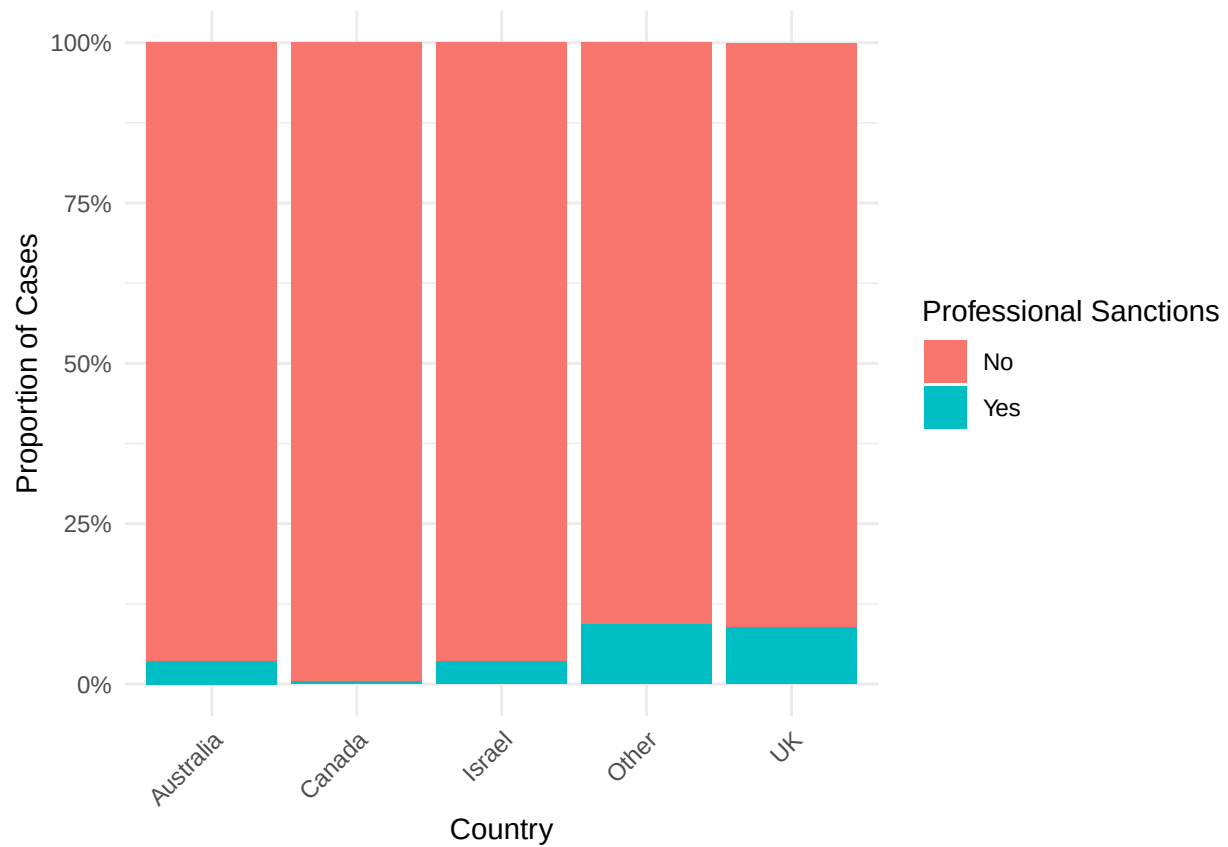
ggplot(hallucination_cases, aes(x = `Legal Field Primary`, fill = `Professional Sanction`)) +
  geom_bar(position = "fill") +
  labs(
    x = "Primary Legal Field",
    y = "Proportion of Cases",
    fill = "Professional Sanctions"
  ) +

```

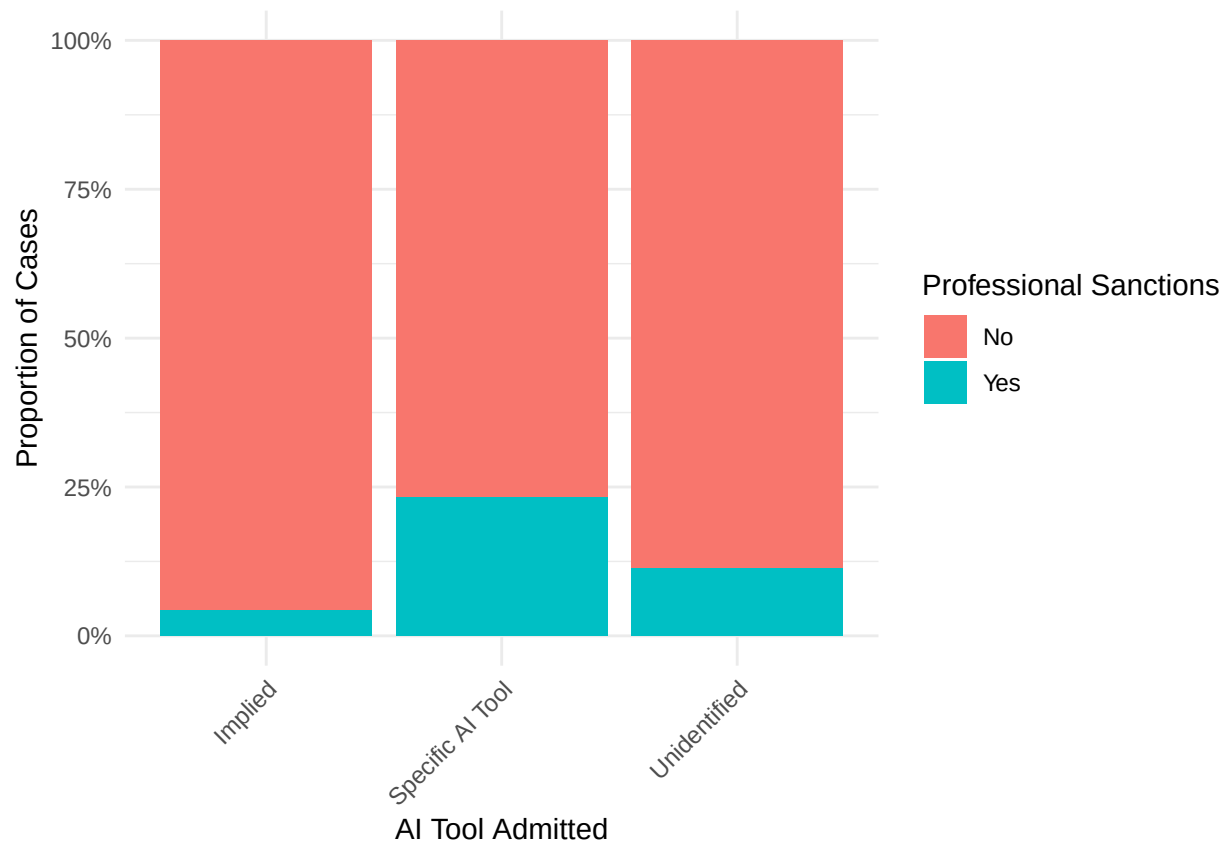
```
scale_y_continuous(labels = scales::percent) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
ggplot(hallucination_cases, aes(x = `country_group`, fill = `Professional Sanction`)) +
  geom_bar(position = "fill") +
  labs(
    x = "Country",
    y = "Proportion of Cases",
    fill = "Professional Sanctions"
  ) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
ggplot(hallucination_cases, aes(x = `ai_tool_group`, fill = `Professional Sanction`)) +  
  geom_bar(position = "fill") +  
  labs(  
    x = "AI Tool Admitted",  
    y = "Proportion of Cases",  
    fill = "Professional Sanctions"  
  ) +  
  scale_y_continuous(labels = scales::percent) +  
  theme_minimal() +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
library(broom)
```

```
## Warning: package 'broom' was built under R version 4.5.2
```

```
#making professional sanctions a binary variable and then running a glm
hallucination_cases <- hallucination_cases %>%
```

```
  mutate(
    professional_sanction_binary = if_else(
      `Professional Sanction` == "Yes",
      1,
      0
    )
  )
```

```
logistic_model <- glm(`professional_sanction_binary` ~ `country_group` + `Legal Field Primary` + `ai_tool_group`,
  data = hallucination_cases, family = "binomial")
summary(logistic_model)
```

```
##
```

```
## Call:
```

```
## glm(formula = professional_sanction_binary ~ country_group +
##     'Legal Field Primary' + ai_tool_group, family = "binomial",
##     data = hallucination_cases)
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -4.2720     0.6604  -6.468 9.91e-11 ***
```

```
## country_groupCanada -1.7454 1.1771 -1.483 0.1381
## country_groupIsrael 0.4959 0.9545 0.520 0.6034
## country_groupOther 1.5383 0.6211 2.477 0.0133 *
## country_groupUK 0.8995 0.7573 1.188 0.2349
## 'Legal Field Primary'bankruptcy 0.1064 0.5602 0.190 0.8494
## 'Legal Field Primary'civil rights -1.1690 0.4617 -2.532 0.0113 *
## 'Legal Field Primary'contract -0.2786 0.3288 -0.847 0.3968
## 'Legal Field Primary'criminal 0.5994 0.4423 1.355 0.1753
## 'Legal Field Primary'employment -0.7810 0.4302 -1.815 0.0695 .
## 'Legal Field Primary'family -0.0666 0.4349 -0.153 0.8783
## 'Legal Field Primary'habeas -13.5470 438.2504 -0.031 0.9753
## 'Legal Field Primary'immigration 0.8429 0.5368 1.570 0.1164
## 'Legal Field Primary'IP -0.1824 0.7116 -0.256 0.7977
## 'Legal Field Primary'landlord-tenant -0.5596 1.0885 -0.514 0.6072
## 'Legal Field Primary'other -0.4804 0.4125 -1.165 0.2441
## 'Legal Field Primary'tax -0.2558 0.8257 -0.310 0.7567
## 'Legal Field Primary'tort -0.5643 0.4146 -1.361 0.1735
## ai_tool_groupSpecific AI Tool 2.1332 0.2417 8.826 < 2e-16 ***
## ai_tool_groupUnidentified 1.2661 0.2272 5.573 2.50e-08 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 949.98 on 1806 degrees of freedom
## Residual deviance: 813.73 on 1787 degrees of freedom
## (258 observations deleted due to missingness)
## AIC: 853.73
##
## Number of Fisher Scoring iterations: 14
```

```
#tidy(logistic_model)
```

Why this model: This model is appropriate for our question of how the other variables in the dataset affect the probability of hallucinated cases receiving a professional sanction because we are testing what the strongest predictor of a professional sanction is between countries with 50 or more cases, primary legal field, and AI tool group. These are relevant explanatory variables because they all test different areas and thus have low chance of multicollinearity. Glm is also suited for this analysis because we made professional sanction into a binary variable.

Interpretation: Based on the model, first we see that Canada, Israel, and the UK don't have a significant difference in predicting professional sanctions. However, countries with 50 or more cases lumped into the "other" category have a significantly higher likelihood to issue professional sanctions in hallucinated cases. Specifically, holding all other predictors fixed, a hallucinated case being in the "other" category of countries increases the log odds of receiving a professional sanction by 1.5383. In terms of legal fields, we again see that most legal fields don't matter in terms of professional sanctions in hallucinated cases, besides civil rights. Holding all other predictors constant, a hallucinated case being in the legal field of civil rights decreases the log odds of receiving a professional sanction by 1.1690. Lastly, we see that AI tools are the strongest predictors based on coefficients and statistical significance. Holding all other predictors fixed, a hallucinated case using specific AI tools has an increased log odds of 2.1332 in terms of receiving a professional sanction and unidentified tools 1.2661.

Limitations: One limitation of our model is that many distinct countries are lumped into the "other" category of a higher likelihood of receiving a professional sanction. Based on the model, all we know is that Canada, Israel, and the UK don't have distinct differences in terms of professional sanctions in hallucinated cases,

but the rest of the world does. This has limited explanatory power when looking at professional sanction by country throughout the world.

```
mean(hallucination_cases$professional_sanction_binary)
```

```
## [1] 0.07845036
```

Another limitation is that out of all hallucination cases, only 7.8 percent have a sanction. When there's a severe imbalance like this in the data, the model can become sensitive to small changes and underestimate the likelihood of “Yes” sanctions as opposed to “No” sanctions.